



BEARING LIFE CALCULATION REPORT

INPUT DATA

Parameter	Value	Symbol	Unit
Type of Bearing	Spherical Roller Bearing		
Number of Rows	1	i	
Ball Diameter	10.0	Dw	mm
Radial Load	10000.0	Fr	N
Axial Load	30000.0	Fa	N
Rotational Speed	500.0	N	rpm
Required Minimum Life	10000.0	Lh	hours

DYNAMIC LOAD CAPACITY CALCULATION

Size factor (f) =

Radial Factor (X) = 0.56

Axial Factor (Y) = 1.0

Life Exponent (p) = 3.0

Equivalent Dynamic Load (P) = $X \cdot Fr + Y \cdot Fa$
= $(0.56 \times 10000.0) + (1.0 \times 30000.0)$
= 35600.0 N

Dynamic Load Rating (C0) = $f \times i \times Dw^2$
= 1×10.0^2
= 1140.0 N

$$\frac{60 \times L_h \times N}{106} = \frac{60 \times 10000.0 \times 500.0}{106}$$

= 300.0 million rev

Dynamic Load Capacity (C) = $P \times (L10)^{3.0}$
= $35600.0 \times 300.0^{3.0}$
= 238318.1302292523 N



BEARING DIMENSIONS

Parameter	Value	Symbol	Unit
Outer Diameter	110.0	D	mm
Inner Diameter	70.0	d	mm
Bearing width	20.0	B	mm
Number of balls	19.0	Z	